

Clustering effect in higher-order moments across various timescales in the cryptocurrency market

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Abstract

This paper examines the risk cascade in high-order moments between investors and traders operating at different investment horizons in the cryptocurrency market. After constructing realized skewness and kurtosis on five major cryptocurrencies (BTC, ETH, XRP, LTC, and XLM) based on high-frequency (5-minute) price data, this paper estimates the transition probability of high or low skewness/kurtosis from long timescale to short timescale using the wavelet hidden Markov tree model. The results show that the negative skewness (low kurtosis) in the long timescale is likely to be followed by the negative skewness (low kurtosis) in the short timescale. However, high kurtosis in the long timescale does not necessarily lead to high kurtosis in the short timescale. These results are robust across various sub-periods and market states. In addition, it is found that in a market with higher market efficiency, high-frequency traders can absorb risk faster and thus show a lower probability of maintaining a high-risk state when a low-frequency trader is exposed to high risk. At the beginning of the COVID-19, the extended analysis indicates, for the case of skewness, the average transition probability of low-to-low and high-to-high across different timescales has increased. These findings indicate the presence of a higher-order moments risk transmission from long-period (low-frequency) investors to short-period (high-frequency) investors, which have implications regarding market efficiency and predictability of higher-order moments across timescales in the debatable cryptocurrency market.

JEL Classification: G11; G12; G13

Keywords: Clustering effect; Higher-order moments cascades; Cryptocurrencies; Wavelet Hidden Markov Tree Model; Market efficiency

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1 Introduction

Under extreme events and turbulent market states, non-zero skewness and excess kurtosis often manifest, substantially deviating returns from the Gaussian distribution¹ and making volatility - the second moment of return distribution - a partial measure of risk. In fact, in the presence of fat-tailed return distribution, extreme risk probability occurs frequently, and downside risk probability is not always equal to upside risk probability. Interestingly, the highly speculative market of cryptocurrencies represents a typical example characterized by frequent periods of extreme booms and busts, which pumps up excess kurtosis and agitates skewness away from zero, ultimately resulting in extreme volatility levels (Kumar et al. 2022). These features are facilitated by the irrational and emotion-driven trading activities of individual investors, under which dramatic price changes and flash crashes are impressive (Ahmed 2022; Bouteska et al. 2022; Naeem et al. 2022). Accordingly, relying on volatility to capture cryptocurrency risk is not necessarily optimal, which suggests the necessity to consider higher-order moments such as return asymmetry (skewness) and excess kurtosis.

With the high risk surrounding the cryptocurrency market, market participants operating at various investment horizons (Bouri et al. 2020; Gradojevic and Tsiakas 2021) are eager to gain a clearer understanding of how (higher-order) risk covering skewness and kurtosis is cascaded among them to make a more prudent trading and investment strategies. While an important study by Gradojevic and Tsiakas 2021 on the volatility cascades in the cryptocurrency market shows that when moving from long to short horizons, high (low) volatility at long horizons is likely to be followed by high (low) volatility at short horizons. This study ignores whether a similar pattern exists in high-order moments such as skewness and kurtosis. Motivated by the suboptimal inferences made under the assumption that volatility is a comprehensive measure of risk when returns are substantially deviated from the Gaussian distribution, the goal of this paper is to examine whether there is a transmission of skewness and kurtosis risk from the long term to the short term and whether this transfer depends on the state of skewness and kurtosis in the cryptocurrency market. In another word, we investigate if skewness/kurtosis is positive (negative)/high (low) in the long term, whether skewness/kurtosis will be positive (negative)/high (low) in the short term with a significantly higher probability. Addressing these

¹Their importance of higher-order moments to asset pricing, portfolio allocation, and risk management cannot be overestimated (e.g., Martellini and Ziemann 2010).

research questions allows us to move the debate on cryptocurrency risk cascades to higher-order moments, which have implications regarding higher-order moments predictability and the trading and investment decisions of various participants in the cryptocurrency market. The emphasis on different timescales is motivated by the fact that different types of traders have different horizons (see, Bouri et al. 2020; Gök et al. 2022). Short-term traders constantly observe the market to reassess their current positions and execute trades with high frequency. Long-term traders may observe the market only once a day or less. For example, a rapid rise in prices followed by a rapid fall of the same magnitude is a big deal for intraday traders, but not for central banks and long-term investors. Long-term traders are only interested in large price movements, which usually occur only over long intervals. Therefore, there is no need for long-term traders with open positions to watch the market all the time. In other words, they judge markets, prices, and risk on a rough grid of time. Long time intervals show the views of long-term traders while short time intervals reflect the views of short-term traders.

Our empirical methodology is based on the wavelet hidden Markov tree (WHMT) model implemented by Gençay et al. (2010). Applying this model, we can extract relevant information about investors with different investment horizons from high-frequency data and further analyze the information flow among them. The first component of this model is the discrete wavelet transform (DWT) ², which has been investigated in several financial and economic time series due to its capacity of covering a number of concepts and features such as non-stationarity, multiresolution and approximate decorrelation that emerge from wavelet filters. It has been applied in the study of the co-movement among international stock market, analysis of the evolution of the oil, commodity price, etc. (Aguiar-Conraria and Soares 2011; Madaleno and Pinho 2014; Shik Lee 2004). The second component of this model is hidden Markov tree (HMT). The hidden Markov model (HMM) is a kind of Markov chain, and its state cannot be directly observed but can be observed through observation vector sequence. Each observation vector is expressed as various states through some probability density distribution, and is generated from a sequence of states with a corresponding probability density distribution. The HMM was successfully applied

²Wavelet tools extend the usual Fourier transform (which is applicable to stationary signals) to non-stationary signals. While Fourier transformation allows only for the decomposition in the frequency space, wavelet allows for a unique time-frequency decomposition and covers the evolution of the signal in two dimensions: time and frequency. Compared to continuous wavelet transform (CWT), DWT is more useful for applying to regular time series analysis with observations at discrete intervals.

to acoustic signal modeling in the 1980s. Besides its use in engineering fields, such as image processing, artificial speech synthesis, seismic exploration, biological signal processing, the HMM has been applied in the field of financial economics, covering volatility in financial markets and stock prices forecasts (Bucci et al. 2021; Hassan and Nath 2005; Nguyen 2018).

Empirically, we look at five major cryptocurrencies, namely Bitcoin (BTC), Ethereum (ETH), Ripple (XRP), Litecoin (LTC), and Stellar Coin (XLM) which constitute almost 85% of the cryptocurrency capitalization. Our main empirical finding is that for cryptocurrencies, positive(negative) skewness in the long term is likely to be followed by positive (negative) skewness in the short term; high (low) kurtosis in the long-term is also likely to be followed by high (low) kurtosis in the short term. However, high kurtosis is less likely to be persistent when moving from long term timescales to the short-term timescales. This is consistent with our expectation that high-frequency investors such as traders can digest risk faster. The above results are robust across different cryptocurrencies and sub-periods.

In our empirical analysis, we investigate the impact of long-term higher-order risk on short-term investors, rather than the reverse - how short-term higher-order risk affects long-term investors. This approach is based on the Heterogeneous Market hypothesis proposed by Müller et al. (1997), which suggests that traders with different investment horizons perceive and react to risks differently. Specifically, short-term investors tend to be more sensitive to short-term fluctuations in the market and may react to changes in long-term higher-order risks that can impact their long-term investment objectives. This can result in short-term fluctuations in the market due to their reaction to changes in long-term higher-order risks. In contrast, long-term investors typically adopt a more passive investment approach and focus on generating consistent returns over a longer period. As such, they may be less concerned with short-term fluctuations and may not monitor the market as frequently as short-term traders. Therefore, short-term high-order risk may not have a significant impact on the investment decisions of long-term investors, who are more concerned with long-term expected risks and returns.

To further illustrate this, consider a value investor who invests in a company based on its intrinsic value and long-term growth potential, rather than short-term fluctuations in the stock price. They may conduct extensive research to assess a company's financial statements, competitive position, and industry trends to determine its long-term prospects. Once they have made an

investment, they may hold the stock for several years or even decades, regardless of short-term market fluctuations. Thus, their investment horizon is much longer than that of a short-term trader, and short-term high-order risk is less likely to impact their decision-making process. Similarly, a pension fund manager responsible for investing the assets of a pension fund may have a long-term investment horizon and may not be as concerned about short-term fluctuations in the market. Their primary focus may be to ensure that the fund has enough assets to meet its long-term obligations to pensioners. As a result, they may invest in a diversified portfolio of stocks, bonds, and other assets that offer long-term growth potential and income streams. They may periodically review the portfolio to ensure that it remains aligned with the fund's investment objectives, but they are less likely to react to short-term market fluctuations or high-order risk.

This paper is related to the cryptocurrency literature examining how risk (e.g., volatility and tail risk) is transmitted across cryptocurrencies (Bouri et al. 2019; Corbet et al. 2018; Hasan et al. 2021; Ji et al. 2019; Yi et al. 2018). Specifically, it is most related to the noteworthy work of Gradojevic and Tsiakas 2021, which covers volatility cascades in cryptocurrency trading, showing that volatility clustering represents an important stylized fact about cryptocurrency volatility, not only in the time domain but also in the frequency domain. We extend their findings from the following threefold, which makes the contributions of this current paper to manifest in three major aspects as follows:

First, we focus on higher-order moments risk rather than volatility. It is well known that cryptocurrencies, like other financial assets, do not conform to a Gaussian distribution, and thus volatility may be proved insufficient to fully explain the risk of cryptocurrency market. Many studies document that higher-order moments risk matters for explaining returns on risky securities (Ahmed and Al Mafrachi 2021; Arditti and Levy 1977; Conrad et al. 2013). For example, Ahmed and Al Mafrachi 2021 use intraday trading data to find that the return rate of cryptocurrencies is affected up to the fourth moment, where skewness and over-skewness can help predict the return rate of the next trading day. However, their study ignores the clustering effect of higher-order moments risk in the time-frequency domain.

Second, we discuss how the above pattern is affected by market efficiency which is extremely important for the relatively newborn and emerging cryptocurrency market. Much research

suggests that cryptocurrencies are in an inefficient market, but are moving toward more efficient (Kristoufek 2018; Urquhart 2016). We use the latest data and perform five formal weak-form Efficient market hypothesis (EMH) tests on five major cryptocurrencies. The empirical finding shows that all of the five cryptocurrencies are in an inefficient market while BTC and ETH show some efficiency as they are larger, better known, and more actively traded. In an efficient market, short-term investors should be able to absorb risk at a faster rate and show less dependence on long-term risk status.

Third, we discuss how the above pattern is being affected by COVID-19, which has caused huge changes in the global economy and financial markets and increased risk and uncertainty in financial markets. As an alternative to fiat currencies, cryptocurrencies gained popularity among investors following the pandemic, mostly motivated by cheap money (arising from expansionary monetary policies by the US Federal Reserve and other major central banks) and the acceleration in the digitalization phenomenon. The pandemic also has changed some important features of the cryptocurrency market. For example, Assaf et al. 2022 show that the long-term memory of most cryptocurrencies declined sharply after the COVID-19 outbreak. Our empirical finding shows that higher-order risk clustering in major cryptocurrencies becomes more pronounced during COVID-19. This finding may help current and potential investors in the cryptocurrency market to better understand how the risk of their investments is affected by other market participants.

In summary, the contribution of our current paper is reflected in the following three aspects. First, this paper adds to the limited literature on higher-order risks in the cryptocurrency market, by highlighting some important features of higher-order risk that distinguish it from the widely watched volatility and tail risk. Second, it discusses how risk contagion patterns in the cryptocurrency market evolve as the market matures. This is particularly important in this newborn cryptocurrency market, which will help traders and investors refine their trading and investment decisions. Third, this paper adds to the literature on the impact of the pandemic on the cryptocurrency market. The findings add to the limited studies in cryptocurrencies on the clustering effect of the skewness and kurtosis of major cryptocurrencies across various timescales, and show evidence on the predictability of higher-order moments across timescales, which have practical implications for portfolio allocation and risk management.

The structure of this paper is as follows. Section 2 gives a brief literature review regarding

the cryptocurrency market. Section 3 describes the data and provides the empirical model. Section 4 presents the empirical results on the clustering effect in higher-order moments of major cryptocurrencies across various timescales, exploring how the main results change with market evolution, and under the COVID-19 shock. Finally, some concluding remarks are offered in Section 5.

2 Literature Review

Over the past five years, the number of academic articles on the finance and economics of cryptocurrencies has been growing exponentially, reflecting the tremendous market growth in this new (digital) asset class, its integration with the financial market, and the increasing attention from the financial media and investors. Our current research is mostly related to three branches of this academic literature. The first branch examines risk spillovers among different cryptocurrencies and between cryptocurrency and other asset classes using a variety of methodologies. The second branch focuses on efficiency in this newborn market and tests the EMH proposed by Fama (1970), which is the most essential theory relating to the modeling of financial assets and well-functioning of financial markets. The third branch investigates the impact of crisis periods, notably the COVID-19 pandemic, on the cryptocurrency market.

In the first branch, many recent studies focus on volatility and tail risk. For example, Kim et al. (2021) constructed a VCRIX³ volatility index to capture investor expectations about the cryptocurrency ecosystem based on the Heterogeneous Auto-Regressive (HAR) model. Borri (2019) estimated the conditional tail-risk in the markets for Bitcoin, Ether, Ripple, and Litecoin and found that these cryptocurrencies are highly exposed to tail-risk within the crypto markets, while they are not exposed to tail-risk with respect to other global conventional assets, such as U.S. equities or gold. Nguyen et al. (2020) examined the network of tail risk spillovers among major cryptocurrencies using LASSO quantile regression, and showed that the right-tail dependence is significantly stronger than the left-tail dependence. Zhang et al. (2021) used several risk metrics such as expected shortfall (ES) and lower partial moment (LPM) to measure

³The VCRIX is a novel cryptocurrency volatility index designed to capture investor expectations regarding the cryptocurrency market. Its function is similar to that of the VIX index, which is widely used in the equity market as a measure of implied volatility for the S&P 500.

downside risk, and found a positive cross-sectional relation between downside risk and future returns in the cryptocurrency market. While these researches have given important insight into the risk in the cryptocurrency market, they have overlooked higher-order moments risk covering skewness and kurtosis despite clear evidence that cryptocurrency returns deviate substantially from the Gaussian distribution and are highly subject to booms and busts. Previous studies have revealed that higher-order moments risk matters in explaining returns on conventional risky financial instruments. Mitton and Vorkink (2007) showed that investors with lottery-like preferences sacrifice mean-variance efficiency to obtain greater levels of positive skewness in their equity portfolios. Kinatder and Papavassiliou (2019) found that realized kurtosis is the most dominant predictor among higher moments and a set of factors of subsequent sovereign bond returns. In the area of cryptocurrencies, Ahmed and Al Mafrachi (2021) documented that all moments up to the fifth order are generally relevant to explaining cryptocurrency returns. Hasan et al. (2021) applied a connectedness approach and show evidence of some linkages among major cryptocurrencies based on their skewness and kurtosis. However, to the best of our knowledge, no research has explored how higher-order moments risk is transmitted among investors with different trading horizons. This paper complements the limited literature on the study of high-order moments in cryptocurrency market, providing the first evidence on the presence of higher-order risk cascade in the cryptocurrency market.

In the second branch, many findings have challenged the efficiency of the cryptocurrency market, indicating the inefficiency of information in many cryptocurrencies, with some recent evidence showing a tendency toward an efficient especially for the largest cryptocurrency - Bitcoin. For example, Urquhart (2016) tested the weak form of EMH in several sample periods and found that the Bitcoin market was inefficient overall but has showed some levels of efficiency in the most recent sample period. Kristoufek (2018) constructed an efficiency index for the Bitcoin market and pointed out that between 2010 and 2017, the Bitcoin market was largely inefficient. Chu et al. (2019) found evidence of the adaptive market hypothesis (AMH), which is a relaxed form of EMH, with respect to the high-frequency markets of Bitcoin and Ether, suggesting that the efficiency of these markets varies over time. The above researches suggest that cryptocurrencies are mostly inefficient but many of them are moving toward a more efficient market. This paper seeks to connect the above two branches of the literature by examining how higher-order risk transfer pattern changes in the five largest-cap cryptocurrencies with different degree of market

efficiency.

In the third branch, many recent studies have examined the impact of the COVID-19 outbreak on the cryptocurrency market. It is well known that COVID-19 has caused huge changes in the global economy and financial markets, bringing the increased uncertainty, fear, and volatility that have brought into financial markets. This issue is important given that Bitcoin, often regarded as a hedging and safe haven asset (Bouri et al. (2020)), has gained increasing popularity around the pandemic period. In this regard, many studies revealed that the pandemic would change some important features of the cryptocurrency market. For example, Raheem (2021) showed that the safe haven hypothesis of Bitcoin fizzled out after the announcement of COVID-19. Maitra et al. (2022) also found that cryptocurrencies cannot provide incremental gains by hedging stock market risk during the COVID-19 pandemic. Their finding implies that the safe haven property of Bitcoin is only limited to calm and tranquil financial market states. Kumar et al. (2022) uncovered a structural change in the connectedness of the cryptocurrency market evolving in 2020 as the market restructures in reaction to the unprecedented monetary injections as a counter to the COVID-19-induced economic standstill. Assaf et al. (2022) indicated that the long-term memory of most cryptocurrencies declined sharply after the COVID-19 outbreak. Our research provides new sight into how COVID-19 affects the cryptocurrency market by discussing the risk cascades pattern during the pandemic period. It shows for skewness that the average transition probability of low-to-low and high-to-high across different timescales has increased under the COVID-19 outbreak period.

To the best of our knowledge, this is the first study on the transmission of skewness and kurtosis clustering across frequencies in the cryptocurrency market. This current study shows for the first time that negative skewness (low kurtosis) in the long timescale tends to be followed by negative skewness (low kurtosis) in the short timescale, whereas high kurtosis does not exhibit a similar clustering effect between the long and short timescale. This can help to further develop and refine our understanding of new aspects of higher-order moments risk cascade in the cryptocurrency market and provide beneficial information for investors and traders operating at various investment horizons.

3 Data and Methodology

Our dataset consists of 5-minute prices of five major cryptocurrencies against the US dollar, namely Bitcoin (BTC), Ethereum (ETH), Ripple (XRP), Litecoin (LTC), and Stellar Coin (XLM), collected from Bifinex, one of the major cryptocurrency exchanges. Notably, these major cryptocurrencies combined have a total market capitalization of around 1.6 trillion US dollars representing a dominance portion of over 85% of the global cryptocurrency capitalization.

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Considering the balance between estimation precision and minimization of white noise, we use 5-minute high-frequency price data to generate 288 observations every day.⁵ Given that at the early days of the cryptocurrency market, before 2017, the cryptocurrency market was not very popular and its trading activities were moderate, our sample period starts from year 2017 around the launch of Bitcoin futures in late 2017. Specifically, the sample data spans the following periods: BTC/USD, ETH/USD, LTC/USD and XRP/USD from November 1, 2017, to October 31, 2022, and XLM/USD from May 1, 2020, to October 31, 2022, according the availability of some cryptocurrencies. [For example, for XLM, the sample period starts from May 1, 2020, which is due to the necessity to ensure that the data length is 2^J , and May 1, 2020 is the first date that satisfies this requirement after XLM started trading.] It is important to note that the length of the data sample is restricted to the observation length 2^J ($J = 1, 2, 3...$), since we employ the discrete wavelet analysis as the main research method in this paper.

⁴These figures are based on data as of December 31, 2021.

⁵We use these price observations to calculate the 5-minute return and daily return for each cryptocurrencies. To calculate the 5-minute return, we subtract the price of the asset at the beginning of a 5-minute period from the price at the end of the period, and then divide the result by the price at the beginning of the period. Mathematically, this can be expressed as $(P_{end} - P_{start})/P_{start}$, where P_{end} is the price of the asset at the end of the 5-minute interval, and P_{start} is the price at the beginning of the interval.

For daily returns of assets that trade continuously, such as cryptocurrencies, a fixed time point is typically chosen to define the end and start periods of a day. For this purpose, the closing price at 4:00 PM of the current day is often used as the end price, while the closing price at 4:00 PM of the previous day is used as the start price, creating a 24-hour period for each trading day.

3.1 Descriptive Statistics

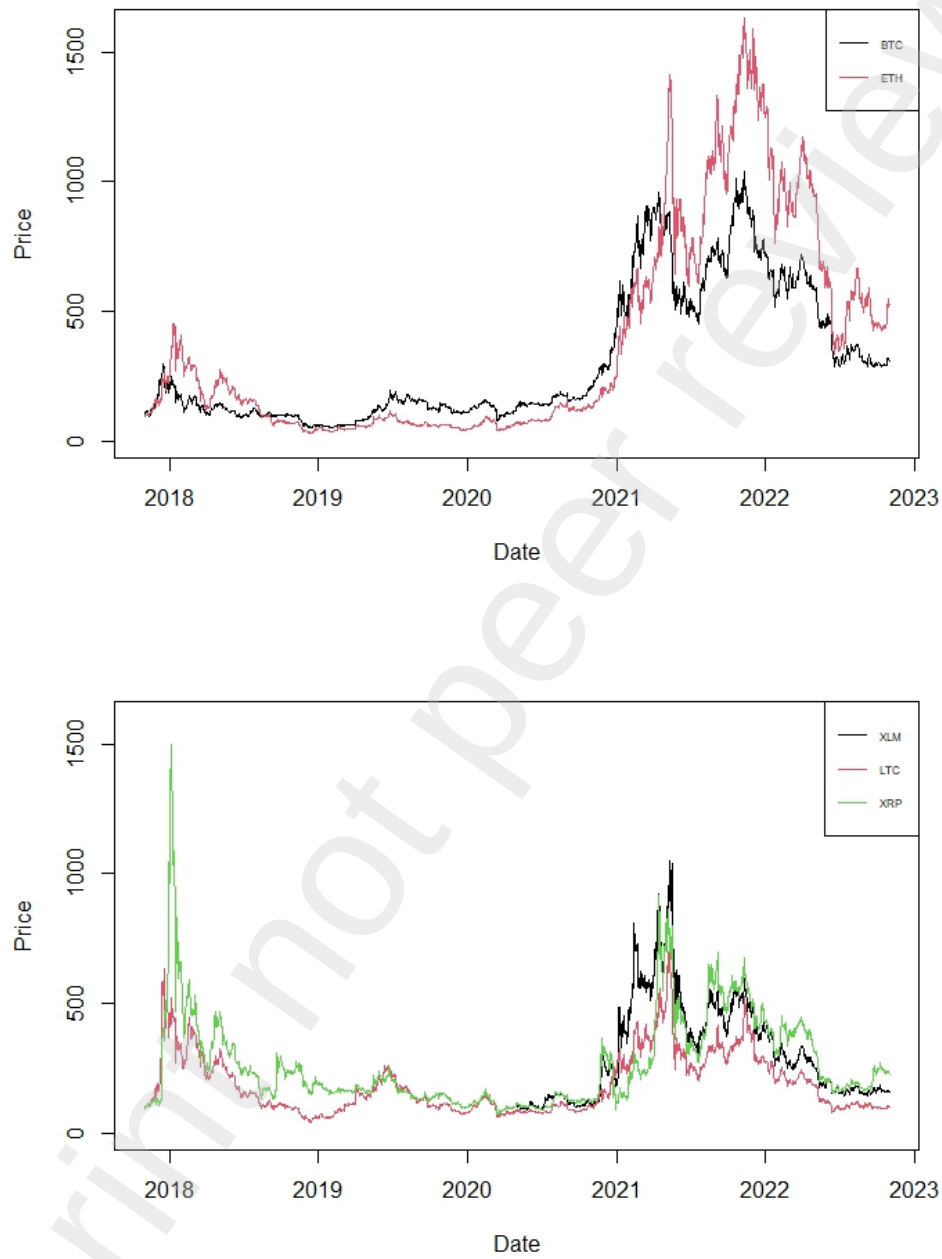
Figure 1 presents a time-series plot of the prices of the five cryptocurrencies described above from November 2017 to October 2022. We standardized the raw data to make it easier to compare the price trends among each cryptocurrency. One of the most remarkable features of the graph is the high degree to which the prices of the five cryptocurrencies move in tandem. From November 2017 to May 2018, the cryptocurrency market experienced a major bull market, with the price of Bitcoin rising from \$4,000 to nearly \$20,000 and the price of ETH rising nearly tenfold. In December 2017, there were some positive developments in the market due to the launch of mainstream cryptocurrency futures contracts, which solidified investor confidence and further improved market liquidity. From May 2018 to the beginning of 2019, cryptocurrency prices declined sharply, mainly due to regulatory policy changes. In early 2020, the global outbreak of COVID-19 caused wild swings in global stock markets, gold, and cryptocurrencies. After that, the prices of cryptocurrencies spiked, especially after October 2020, since the stimulative monetary policies introduced by many central banks devalued fiat currencies and funds flowed into the cryptocurrency market.

Table 1 presents the main descriptive statistics for the daily returns of the five cryptocurrencies in the sample. Interestingly, Bitcoin and Ethereum have relatively smaller standard deviations of around 4.1% and 5.2%, respectively. However, LTC has the greatest left-tail risk, with its extremely low daily return of -37.5%, while XLM is the cryptocurrency with the highest degree of right-tail risk, achieving an extremely high daily return of almost 60%.

The main focus of the paper is the third-order moment skewness and fourth-order moment kurtosis, which represent higher-order risk indicators. The skewness of BTC and ETH returns is negative, indicating that the left side of the return distribution contains more extreme values than the right side, and traders and investors thus should focus more on the left-tail risk of these cryptocurrencies. However, the skewness of LTC, XLM, and XRP is positive, which indicates that the probability of extreme positive returns is large.

Previous studies have confirmed that the return of most stock indexes is featured by negative skewness. The most well-known explanation involves the leverage effect. According to this theory, when bad news arrives in the market, the stock price of the firm under consideration falls and its market value declines. The firm might have to increase the operating leverage,

Figure 1: The daily price for five major cryptocurrencies



Note: The upper panel demonstrates the evolution of the BTC/USD and ETH/USD price data on Bifinex from November 2017, to October 2022. The lower panel demonstrates the evolution of the XLM/USD, LTC/USD and XRP/USD price data on Bifinex from November 2017, to October 2022.

Table I: Descriptive statistics

Names	Sample period	Mean	SD	Min	5% VaR	95% VaR	Max	Skewness	Kurtosis
BTC	2017/11/1-2022/10/31	0.001	0.041	-0.239	-0.066	0.065	0.233	-0.298	4.31
ETH	2017/11/1-2022/10/31	0.001	0.052	-0.324	-0.084	0.082	0.285	-0.350	4.549
LTC	2017/11/1-2022/10/31	0	0.058	-0.375	-0.087	0.086	0.542	0.131	9.23
XLM	2020/5/1-2022/10/31	0.001	0.061	-0.332	-0.084	0.083	0.592	1.244	16.633
XRP	2017/11/1-2022/10/31	0	0.064	-0.515	-0.086	0.089	0.569	0.752	14.714

Note: This table presents summary statistics for daily returns of 5 major cryptocurrencies.

which further increases the probability of the stock price decline. However, this explanation is not suitable for the cryptocurrency market. The relatively mature BTC and ETH markets show negative skewness similar to stock markets, while the other three cryptocurrencies with smaller market caps show positive skewness. One possible explanation for the positive skewness of the three relatively small cryptocurrencies in our sample is the speculative appetite of investors. Cryptocurrencies have shown their value to investors in recent years as BTC prices have risen sharply. However, as the BTC price exceeds \$20,000, retail investors with a speculative appetite are more willing to believe that other cryptocurrencies with lower prices and smaller market capitalization will bring them huge gains. Driven by this psychology, a rise in the price of a small cryptocurrency will further attract more investors into the market, which further increases the probability of future price increases, accentuated by the fear of missing out (Baur and Dimpfl (2018)). As a result, their returns show positive skewness. This can be consistent with the evidence from Table 1 showing that the last three cryptocurrencies all had maximum daily returns of over 50% during our sample period, almost double the maximum daily returns of BTC.

The kurtosis of returns for all cryptocurrencies is higher than 3. This suggests that investors should be aware of the bigger probability of extreme returns from investing in the cryptocurrency market, compared to the Gaussian distribution assumption in the classical asset pricing literature. It is interesting to note that BTC and ETH have significantly lower kurtosis than the other three cryptocurrencies. To some extent, this indicates that the return distribution of cryptocurrencies with high market attention and high trading volume is closer to the normal distribution and potentially their degree of market efficiency is higher.

3.2 Methodology

3.2.1 Higher-order Moments Measures

This paper focuses on higher-order moments risk indicators, skewness, and kurtosis. Constructing skewness and kurtosis using high-frequency data has not been extensively studied until the last few years. According to Amaya et al. (2015), realized third moment (RTM) and realized fourth moment (RFM) can be written in the following form:

$$RTM_t = \sum_{j=1}^{1/\Delta} (r_{j,t})^3 \xrightarrow{p} \sum_{0 < s \leq t} (\Delta p_s)^3,$$
$$RFM_t = \sum_{j=1}^{1/\Delta} (r_{j,t})^4 \xrightarrow{p} \sum_{0 < s \leq t} (\Delta p_s)^4,$$

where $r_{j,t}$ is the change in the log-price level at interval t , defined as $r_{j,t} = p_{j,t} - p_{j-1,t}$.

To be consistent with the previous concepts of skewness and kurtosis risk, we constructed the following indexes of realized skewness (RSK) and realized kurtosis (RKU) with reference to Amaya et al. (2015) :

$$RSK_t = \frac{\sqrt{1/\Delta \sum_{j=1}^{1/\Delta} r_{j,t}^3}}{\left(\sum_{j=1}^{1/\Delta} r_{j,t}^2\right)^{3/2}},$$
$$RKU_t = \frac{1/\Delta \sum_{j=1}^{1/\Delta} r_{j,t}^4}{\left(\sum_{j=1}^{1/\Delta} r_{j,t}^2\right)^2}.$$

The formulas above represent two measures of the distribution of returns, realized skewness and realized kurtosis, which are calculated based on the historical data of asset returns. Realized skewness (RSK) measures the degree of asymmetry in the distribution of returns over a given period, which is represented by the third moment of the distribution. Realized kurtosis (RKU), on the other hand, measures the degree of peakedness or flatness in the distribution of returns, which is represented by the fourth moment of the distribution. Both RSK and RKU are normalized measures, of which higher values indicate greater skewness and kurtosis respectively.

3.2.2 Empirical Model

The empirical analysis in this paper draws on the wavelet hidden Markov tree (WHMT) model proposed by Gençay et al. (2010), which combines discrete wavelet transform (DWT) and hidden Markov tree (HMT). Compared with previous research on time series applying non-linear dependencies via the quantile method, this WHMT can reveal statistics under different time scales (sampling frequency), and further helps us answering how high-order risk faced by low-frequency traders will affect the high-order risk faced by the high-frequency traders.

The first part of the model is DWT. After applying the discrete wavelet transform, the fluctuations of typical time series are concentrated on a few wavelet coefficients, and most wavelet coefficients are very small. Therefore, the following Gaussian mixture model (GMM) can be used to represent the distribution of wavelet coefficients. The second part of the model is the implicit markov tree. Assume that each wavelet coefficient is associated with implied condition $S \in \{1, 2, \dots, M\}$, and the probability of each state is $P_S(m)$, for each state w . Wavelet coefficients under state $S = m$ obey Gaussian distribution $N(u_m, \sigma_m^2)$, so M state Gaussian mixture model can be expressed as

$$f_w(w) = \sum_{m=1}^M P_S(m) f_{w|S}(w | S = m),$$

where $f_{w|S}(w | S = m) = \frac{1}{\sqrt{2\pi\sigma_m^2}} \exp\left(-\frac{(w-u_m)^2}{2\sigma_m^2}\right) = g(w, u_m, \sigma_m^2)$. It is worth noting that the probability density of the wavelet coefficients is conditionally Gaussian, but in essence it is non-Gaussian.

DWT is well known for its ability to decorrelate signals. However, there still exist high-order correlations between wavelet coefficients, as evidenced by the similarity between coefficients across different scales. This correlation can be captured by the HMT structure, as depicted in Figure 3-2 for one-dimensional signals in the wavelet domain. In this structure, the black dots represent wavelet coefficients, and the white dots represent hidden states. The tree has a typical binary structure, where each tree is assumed to be independent and the root node is the coarsest-layer wavelet coefficient.

For each node, denoted by index i , its parent and child nodes are denoted by $\rho(i)$ and $c(i)$, respectively. The relationship between the parent state and its child nodes are captured by the state transition probability $\varepsilon_{i,p(i)}^{m,r}$, which satisfies the following properties: (1) given the state

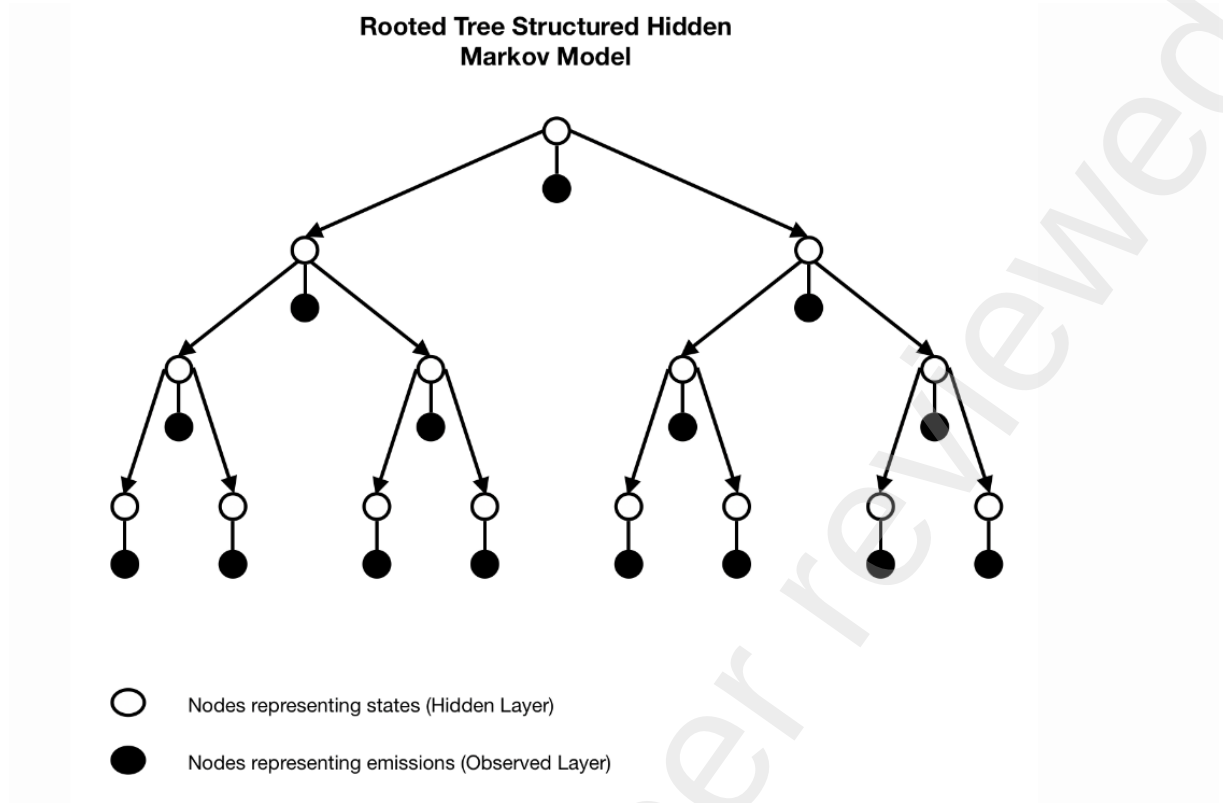


Figure 2: Structure of WHMT

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S_i of a node, the wavelet coefficient w_i at that node is independent of other nodes in the tree; (2) given the state of a node, the states of its parent and child nodes are independent of each other; and (3) given the state of a specific node's parent $S_{\rho(i)}$ and the state of all its child nodes $S_j \in c(i)$, the node (S_i, w_i) is independent of other nodes in the tree. For convenience, we define the set of indices for all nodes as I , and denote the set of wavelet coefficients and states as $W = w_i \mid i \in I$ and $S = S_i \mid i \in I$, respectively.

The model parameters for the wavelet-based hidden Markov tree (WHMT) can be summarized as follows:

- The probability distribution of the root node, denoted by $P_{s_1}(m)$, where m ranges from 1 to M .
- The conditional probability, denoted by $\varepsilon_{i,p(i)}^{m,r}$, represents the probability of observing the state m at node S_i given that its parent node $S_{P(i)}$ is in state r . Here, $S_{\rho(i)}$ denotes the

parent node of S_i , and r and m range from 1 to M .

- For each node S_i and state m , two parameters regarding its Gaussian distribution, namely, the mean $u_{i,m}$ and variance $\sigma_{i,m}^2$ of the wavelet coefficients w_i are estimated.

Collectively, the parameter vector can be expressed as $\theta = \{P_{s_i}(m), \varepsilon_{i,p(i)}^{m,r}, u_{i,m}, \sigma_{i,m}^2, m, r \in 1, \dots, M, i \in \mathbb{I}\}$. In the empirical analysis following, our primary focus is on the transition probability $\varepsilon_{i,p(i)}^{m,r}$, which captures the transmission of high-order risk from low-frequency dimensions to high-frequency dimensions. The estimation of this probability is of particular importance, as it provides insights into the underlying structure of the wavelet-based hidden Markov tree (WHMT) and the clustering effect of high-order risk in the cryptocurrency market.

4 Empirical Results

4.1 Test of Market Efficiency

One of the most important principles for the proper functioning of financial markets is market efficiency, which has been found to be prominent in the equities of well-established markets. However, the emergence of the cryptocurrency market is a recent phenomenon, and its trading volume is comparably lower than traditional assets such as equities and bonds. As a result, the efficiency of the cryptocurrency market may not match that of well-established asset markets. Nonetheless, the cryptocurrency market has witnessed significant development, with the launch of futures contracts on Bitcoin and Ethereum, and the growing attention of investors. The market is approaching mainstream and developing towards in a more and more efficient direction.

We first examine the market efficiency of each cryptocurrency under study before further discussing the parameter estimation results of the WHMT model. The main reason why we focus on market efficiency is that the dependence of realized skewness and realized kurtosis across different time scales can be affected by market efficiency. In a completely efficient market, investors cannot predict the risks that high-frequency trading will face from the current risk state of low-frequency trading. We will further verify this in the later empirical part.

Table 2 reports the results of market efficiency tests (namely Ljung-Box test, Runs test, Bartels

test, automatic variance ratio (AVR) test, and BDS (Brock-Dechert-Scheinkman) test ⁷ of the cryptocurrencies under study over the full sample period. It can be seen that for the 5 cryptocurrencies, the return series are not i.i.d. Compared to LTC and XLM, BTC, ETH, and XRP show a higher degree of market efficiency, for which we cannot reject that the return series over the sample period is independent. This result can be due to the relatively higher market value and trading volume of BTC, ETH, and XRP.

Table II: Market efficiency test for the 5 mainstream cryptocurrencies over the full sample period

Coin	Sample period	L-B test	Runs test	Bartels test	AVR test	BDS test
BTC	2017/11/1-2022/10/31	0.632	0.232	0.218	0.986	0.000
ETH	2017/11/1-2022/10/31	0.538	0.267	0.224	0.785	0.000
XRP	2017/11/1-2022/10/31	0.567	0.298	0.239	0.635	0.000
LTC	2017/11/1-2022/10/31	0.321	0.000	0.000	0.000	0.000
XLM	2020/5/1-2022/10/31	0.333	0.000	0.000	0.000	0.000

Note: Ljung-Box (L-B) is for testing auto-correlation, Runs test and Bartels test are for testing independence, Bartels test is for testing automatic variance ratio (AVR) is for testing random walk process, and Brock-Dechert-Scheinkman (BDS) is for testing i.i.d. process. The detailed explanations for these tests are shown in footnote 6.

Table 3 reports the P-values of some relevant efficiency tests across different sample periods of BTC. Overall, we cannot reject the null hypothesis of weak market efficiency for larger P values.

⁷The choice of these tests is inspired from the related literature.

The Ljung-Box test, Runs test, Bartels test, Automatic Variance Ratio (AVR) test, and BDS (Brock-Dechert-Scheinkman) test are widely used statistical tests for examining various characteristics of financial time series data, such as autocorrelation, periodicity, long memory, independence, and runs. These tests play a crucial role in testing market (in)efficiency, as they can reveal patterns in data that suggest market inefficiencies, thereby helping investors better understand the underlying factors that drive market movements and identify potential opportunities for profit. In particular, The Ljung-Box test detects the presence of autocorrelation, which measures the degree of correlation between successive data points. The Bartels test detects periodicity, which measures the regularity in the data that may suggest seasonality. The AVR test examines the constancy of variance over time, while the BDS test detects long memory or dependence in financial time series data. The Runs test examines the occurrence of runs in the data, which are consecutive periods where the price moves in the same direction. By analyzing these different characteristics, these tests provide valuable insights that can help market participants make better-informed decisions about financial assets, and they can also be used to test the (in)efficiency of financial markets.

The second row of Table 3 gives the specific null hypothesis for each test. For the full sample period (2017/11/1-2022/10/31), 3 of the 5 tests support weak market efficiency; For the earliest year of the sample period (2017/11/1- 2019/10/31), 3 of the 5 tests reject the null hypothesis of weak market efficiency. This shows that market efficiency is low in the early stage of market development. For the second half of the sample (2019/11/1-2022/10/31), 4 of the 5 tests support weak market efficiency. This shows that over time, the market trading mechanism has gradually improved, the market liquidity has become higher, and the market efficiency of the BTC market has gradually improved. This is consistent with Urquhart 2016's finding that Bitcoin is in an inefficient market but may be in the process of moving towards an efficient market, as well as with the findings of Tiwari et al. 2018 and Aslan and Sensoy 2020. It should be noted that the P-value of the BDS test is 0 irrespective of the sample period. This shows that no matter which sub-period is used, the return series of BTC is not i.i.d. In fact, the return series is not even independent in the first half of the sample period, while the assumption that returns are independent has been supported in the second half of the sample period based on Runs and Bartels tests.

Table III: Efficiency test in BTC market across different sample periods

Test	Ljung-Box test	Runs test	Bartels test	AVR test	BDS test
2017/11/1-2022/10/31	0.157	0.985	0.008	0.866	0.000
2017/11/1-2019/10/31	0.671	0.706	0.236	0.708	0.000
2019/11/1-2022/10/31	0.323	0.000	0.000	0.704	0.000

Note: Ljung-Box (L-B) is for testing auto-correlation, Runs test and Bartels test are for testing independence, Bartels test is for testing automatic variance ratio (AVR) is for testing random walk process, and Brock-Dechert-Scheinkman (BDS) is for testing i.i.d. process. The detailed explanations for these tests are shown in footnote 6.

4.2 Dependence of Skewness and Kurtosis across Time Scales

Our empirical analysis focuses on estimating the transition probabilities of higher- order risk skewness and kurtosis from long-term (low-frequencies) to short-term (high-frequencies) by applying the WHMT model.

4.2.1 The Case of Bitcoin

Table 4 presents the transition probabilities of BTC realized skewness across different timescales for the full sample period. The values reported in the table represent the probability of a particular skewness state (high or low) in timescale j , given the state in timescale $j + 1$. The transition probabilities provide insights into the vertical dependence and clustering effect of cryptocurrency skewness and how this dependence varies in different market states. For timescales $j = 4$ and higher, the negative-to-negative transition probability significantly exceeds 50%, indicating that negative skewness tends to persist for almost all transaction periods as long as it is not too long. However, the dependence of skewness risk is not as apparent in lower timescales. Additionally, for scales $j = 5$ and above, the transition probability from positive to positive is also higher than 50%, suggesting that positive skewness also tends to persist over higher frequency and shorter time periods. These findings provide empirical evidence that skewness clusters among investors with different trading horizons. We then analyze the heterogeneity of this cluster effect under different market states (positive skewness or negative skewness). The results in Table 4 show that the vertical dependence of skewness risk is stronger in the state of negative skewness, as the average probability of transitioning from a low-to-low skewness state across different timescales is larger than that of high-to-high.

One possible theoretical explanation for the observed phenomenon of stronger vertical dependence of skewness risk in the state of negative skewness is asymmetry aversion. This concept refers to the tendency of individuals to prefer outcomes that are more symmetric over those that are more skewed, as individuals assign greater weight to the tails of the distribution, leading to higher perceived risk in situations with negative skewness. In the context of financial markets, negative skewness can result in more pronounced losses in the left tail of the distribution, which can have a greater impact on investor welfare. Therefore, investors may be more averse to negative skewness, leading to a stronger vertical dependence of skewness risk in the low-to-low state, where negative skewness is prevalent. Conversely, in the high-to-high state, where the distribution is more symmetric, the impact of asymmetry aversion may be less pronounced, leading to a weaker vertical dependence of skewness risk.

Another potential explanation is related to herding behavior among investors. During times of high uncertainty or market stress, there may be a tendency for investors to follow the actions

of others rather than making independent decisions. This can result in greater clustering of low skewness states during periods of negative skewness, which can amplify the effects of vertical dependence in skewness risk. This clustering effect could arise due to investors' shared perceptions of risk and the resulting tendency to move in the same direction. Therefore, the impact of negative skewness on investor behavior could be magnified, leading to a stronger vertical dependence of skewness risk in the state of negative skewness.

Table IV: Skewness transition probability across different time scales in BTC market

Scale	negative- to- negative	negative- to- positive	positive- to- positive	positive- to- negative
1	0.5000	0.5000	0.5000	0.5000
2	0.5000	0.5000	0.5000	0.5000
3	0.5000	0.5000	0.5000	0.5000
4	0.5324	0.4676	0.5000	0.5000
5	0.5246	0.4754	0.5123	0.4877
6	0.5687	0.4313	0.5234	0.4766
7	0.6235	0.3765	0.5368	0.4632
8	0.6430	0.3570	0.5689	0.4311
9	0.6659	0.3341	0.6123	0.3877
10	0.7123	0.2877	0.6234	0.3766
11	0.7234	0.2766	0.6758	0.3242
12	0.7356	0.2644	0.6859	0.3141
13	0.7652	0.2348	0.7134	0.2866
14	0.7563	0.2437	0.7456	0.2544
15	0.8120	0.1880	0.7328	0.2672
16	0.8234	0.1766	0.8129	0.1871
17	0.8536	0.1464	0.8345	0.1655
18	0.9123	0.0877	0.9021	0.0979
19	0.9234	0.0766	0.9123	0.0877
Average	0.6882	0.3118	0.6522	0.3478

Note: This table reports the transition probabilities for BTC/USD skewness for the full sample. The reported values are the probabilities of a given skewness state in scale j conditional on the skewness state in scale $j + 1$.

Table 5 reports the transition probabilities for BTC/USD kurtosis across different time scales for the full sample. The probabilities shown indicate the likelihood of a particular kurtosis state in time scale j , given the kurtosis state in time scale $j + 1$. The results reveal that for scales

with $j = 5$ and higher, the kurtosis transition probability of low to low is substantially greater than 50%. However, as observed in Table 5, the dependence of kurtosis risk at a lower time scale is not discernible. Moreover, for scales with $j = 5$ and above, the probability of high to high is also greater than 50%. These findings suggest that the vertical dependence of kurtosis risk is stronger in higher-frequency traders.

The observed vertical dependence of kurtosis risk in higher frequency domains can be explained by various theoretical concepts in financial markets. One such concept is information asymmetry, which occurs when one party in a transaction possesses more or superior information than the other party. Traders who operate in higher frequency domains may have access to more timely and accurate information than those who trade less frequently, resulting in greater clustering of kurtosis states in these higher frequency domains, as traders with better information move in the same direction while these less informed investors may just follow their initial trading strategy and keep their initial position.

Another theoretical explanation for the observed dependence of kurtosis risk in higher frequency domains is herding behavior, which refers to the tendency of traders to follow the actions of others rather than making independent decisions as we discussed above. This behavior can lead to greater clustering of kurtosis states in higher frequency domains, as traders follow the lead of others in their decision-making. The impact of herding behavior on kurtosis risk may be particularly pronounced in higher frequency domains, where traders may be more susceptible to peer influence.

A comparison of the results from Tables 4 and 5 indicates that both tables reveal similar trends in the dependence structure across different time scales. Specifically, Table 4 highlights a stronger dependence of skewness risk in the higher frequency domain, whereas Table 5 reveals a stronger dependence of kurtosis risk in the higher frequency domain. These results provide valuable insights into the dynamics of higher-order risk in BTC/USD returns at different time scales. The findings suggest that investors with different trading horizons may encounter varying levels of higher-order risk passed on by other traders, and therefore, should tailor their risk management strategies accordingly. Thus, understanding the mechanisms that drive the dynamics of higher-order risk can be valuable for risk management and investment decision-making.

Table V: Kurtosis transition probability across different time scales in BTC market

Scale	low-to-low	low-to-high	high-to-high	high-to-low
1	0.5000	0.5000	0.5000	0.5000
2	0.5000	0.5000	0.5000	0.5000
3	0.5000	0.5000	0.5000	0.5000
4	0.5000	0.5000	0.5000	0.5000
5	0.5563	0.4437	0.5000	0.5000
6	0.5423	0.4577	0.5123	0.4877
7	0.6782	0.3218	0.5463	0.4537
8	0.7342	0.2658	0.6123	0.3877
9	0.7564	0.2436	0.6234	0.3766
10	0.8032	0.1968	0.6564	0.3436
11	0.8142	0.1858	0.6863	0.3137
12	0.7985	0.2015	0.7012	0.2988
13	0.8342	0.1658	0.7324	0.2676
14	0.8673	0.1327	0.7568	0.2432
15	0.8972	0.1028	0.7869	0.2131
16	0.9124	0.0876	0.7732	0.2268
17	0.8932	0.1068	0.7754	0.2246
18	0.9542	0.0458	0.8363	0.1637
19	0.9425	0.0575	0.8954	0.1046
Average	0.7360	0.2640	0.6523	0.3477

Note: This table reports the transition probabilities for BTC/USD Kurtosis for the full sample. The reported values are the probabilities of a given Kurtosis state in scale j conditional on the Kurtosis state in scale $j + 1$.

4.2.2 Other Major Cryptocurrencies

For other major cryptocurrencies, the dependence of skewness and kurtosis risk across different timescales is similar to that of BTC.⁸ What we are interested in this part is the relationship between this vertical dependence across different timescales and market efficiency we discussed in section 4.1.

Table 6 reports the average skewness transition probability across different timescales for the full sample period and for each cryptocurrency. We notice that the vertical dependence of the

⁸We would ignore the detailed discussions here, see the appendix for details if you are interested

last two cryptocurrencies (XRP and XLM) is significantly stronger than that of the first three cryptocurrencies (BTC, ETH, LTC). Moving to Table 7, the average kurtosis transition probability across different timescales for the full sample period and for each cryptocurrency shows that the vertical dependence of the last two cryptocurrencies XRP and XLM is significantly stronger than that of the first three cryptocurrencies too.

A reasonable explanation for this interesting finding is that compared with the last two currencies XRP and XLM, the first three currencies have relatively a longer development time⁹, a higher degree of market efficiency, and a return series closer to the independent and identical distribution (In fact, for independent and identically distributed time series, there is no such vertical dependency across different time scales.). The economic institution behind this result is that in more mature markets, high-frequency traders can absorb market risk faster and show less dependence on the risks faced by low-frequency traders. We further argue this in the next paragraph.

High-frequency traders (HFTs) are known to have the ability to quickly absorb market risks. For instance, Brogaard et al. (2014) found that HFTs are able to absorb risk faster than non-HFTs in the US equity market, especially during times of market stress. In more mature markets,

⁹According to available data, it can be argued that Litecoin (LTC) is more popular than Ripple (XRP) in terms of user adoption and market capitalization. Firstly, Litecoin was launched in 2011, almost two years before Ripple was introduced in 2013. Therefore, Litecoin has had more time to establish itself in the cryptocurrency market and build a strong user base. Secondly, Litecoin has a larger market capitalization compared to Ripple. As of February 2023, Litecoin's market capitalization is around \$10 billion USD, while Ripple's market capitalization is around \$5 billion USD. This indicates that more investors are willing to invest in Litecoin, which could be a result of its higher degree of acceptance among merchants and businesses. Furthermore, Litecoin is considered to be more decentralized compared to Ripple. While Ripple is primarily used by financial institutions for cross-border payments, Litecoin is open to everyone and can be used for a variety of purposes, including peer-to-peer transactions and purchasing goods and services. In terms of daily trading volume, Litecoin also outperforms Ripple. According to CoinMarketCap, as of February 2023, the daily trading volume of Litecoin is around \$3 billion USD, while Ripple's daily trading volume is around \$1 billion USD. This suggests that Litecoin is more actively traded and has a higher degree of liquidity compared to Ripple. Finally, it's worth noting that Litecoin has been widely accepted by online merchants, particularly in the gaming industry, and is a popular payment method for purchasing virtual goods and services. This further demonstrates the wider acceptance and popularity of Litecoin among both consumers and merchants.

In conclusion, based on the available data, it can be argued that Litecoin is more popular than Ripple in terms of user adoption, market capitalization, decentralization, daily trading volume, and merchant acceptance.

where there is greater trading activity, liquidity, efficiency and lower bid-ask spreads, HFTs are able to absorb risks even more quickly. In mature markets, high-frequency traders can execute trades quickly and efficiently, taking advantage of small price discrepancies and maintaining market stability. Hasbrouck and Saar (2013) found that high-frequency traders contribute to market liquidity, particularly in more mature markets with higher trading volumes, and that their activities can help reduce volatility and absorb shocks to the market.

Table VI: Average skewness transition probabilities for the 5 major cryptocurrencies

Coin	negative- to- negative	negative- to-positive	positive- to-positive	positive- to- negative
BTC	0.6882	0.3118	0.6522	0.3478
ETH	0.6932	0.3068	0.6754	0.3246
LTC	0.7234	0.2766	0.7134	0.2866
XRP	0.7653	0.2347	0.7324	0.2676
XLM	0.7723	0.2277	0.7453	0.2547

Note: This table reports the average skewness transition probability across different time scales for the full sample period and each cryptocurrency.

Table VII: Average kurtosis transition probabilities for the 5 major cryptocurrencies

Coin	low-to-low	low-to- high	high-to- high	high-to- low
BTC	0.7360	0.2640	0.6523	0.3477
ETH	0.7534	0.2466	0.6754	0.3246
LTC	0.7435	0.2565	0.7023	0.2977
XRP	0.7892	0.2108	0.7534	0.2466
XLM	0.7963	0.2037	0.7453	0.2547

Note: This table reports the average kurtosis transition probability across different time scales for the full sample period and each cryptocurrency.

Among existing studies highly relevant to our current study, Qiao et al. (2020) applied wavelet coherence to investigate the return and volatility comovement between Bitcoin and other cryptocurrencies. They found that Bitcoin has a significant impact on other cryptocurrencies and is ahead of other cryptocurrencies. However, their comovement analysis is confined to return and volatility levels. Moreover, their sample interval spans between April 21, 2014 and October

12, 2019 thus missing the COVID-19 pandemic and the Russia- Ukraine war periods. Naeem et al. (2021) analyzed the time-frequency interdependence of cryptocurrency market and found the short and long-run market integration. The return interdependence oscillates at high frequencies while it remains constant at low frequencies. However, their analysis is also limited to the return level. Our study addresses this research gap by examining the higher-order moments dependence in the timescales across different cryptocurrencies. The higher-moment risk transmission investigation provides more targeted and fresh references in the decision-making process for crypto traders and investors operating at various investment horizons. Furthermore, the higher-order moments risk transmission effects among cryptocurrencies help to shed some new light on time-frequency analysis among other financial markets.¹⁰

5 Further Discussion

5.1 Results Since the Outbreak of COVID-19

Given the importance of the pandemic to the dynamics of cryptocurrency risk, we give a special focus to the COVID-19 outbreak sub-period. Table 8 shows the descriptive statistics of the daily returns of the five mainstream cryptocurrencies studied since the outbreak of COVID-19. The average daily return of each cryptocurrency is slightly higher than the average return shown for the full sample (see Table 1), which is consistent with the price trend in Figure 1. Notably, the outbreak of COVID-19 has brought large price movements to the cryptocurrency market and thus an increase in its risk. Specifically, the absolute values of skewness and kurtosis of BTC returns are larger during the pandemic sub-period. The degree of negative skewness has increased, reflecting an increase in the level of market tail risk. Furthermore, the kurtosis has increased, suggesting that an increase in the probability of the extreme value of the returns. Overall, during the COVID-19 outbreak, the distribution of the market returns deviated away from the Gaussian distribution, which further motivates the relevance of considering higher-order moments to reflect a comprehensive view of the risk in the cryptocurrency market.

Table 9 shows the transition probability of skewness since the outbreak of COVID- 19. Compared

¹⁰Bouri et al. (2022) examine the timescale co-movement between Bitcoin and the US stock markets and report evidence of significant co-movement that does beyond the volatility to cover skewness and kurtosis.

Table VIII: Descriptive statistics - since the outbreak of COVID-19

Names	Sample period	Mean	SD	Min	5% VaR	Median	95% VaR	Max	Skewness	Kurtosis
BTC	2020/4/30-2022/10/31	0.004	0.041	-0.239	-0.036	0.003	0.048	0.156	-0.743	5.992
ETH	2020/4/30-2022/10/31	0.006	0.051	-0.314	-0.045	0.004	0.064	0.197	-0.796	5.993
LTC	2020/4/30-2022/10/31	0.003	0.057	-0.335	-0.056	0.003	0.066	0.175	-0.968	5.285
XLM	2020/4/30-2022/10/31	0.005	0.071	-0.332	-0.058	0.002	0.072	0.592	1.720	16.057
XRP	2020/4/30-2022/10/31	0.004	0.074	-0.515	-0.054	0.001	0.067	0.322	-0.441	11.493

Note: This table presents the summary statistics for returns of 5 major cryptocurrencies since the outbreak of COVID-19.

with the results of the full sample in Table 4, the transition probability from high to high across different time scales increases, which is due to the lower market efficiency and more irrational trading during the pandemic. Table 10 shows the transition probability of kurtosis since the outbreak of COVID-19. Compared with the results of the full sample (See Table 5), the average transition probability of low-to-low and high-to-high across different time scales also increases for kurtosis. In the early days of the pandemic, lockdown policies caused economic downturns in many countries. To stabilize the US economy, the Federal Reserve implemented a series of expansionary monetary policies. Investors anticipated that fiat money will depreciate and thus investment in non-sovereign and decentralized assets such as cryptocurrencies to hedge this risk becomes a major theme in the financial markets. This is supported by previous studies that provide evidence of the hedging role of Bitcoin against uncertainty (Demir et al. 2020; Fang et al. 2019; Goodell and Goutte 2021). For example, Demir et al. (2020) used the wavelet coherence analysis to show that there is a negative relationship between Bitcoin value and the number of reported cases and deaths. Their finding indicates the hedging role of cryptocurrencies against the uncertainty raised by COVID-19. However, the risks in the cryptocurrency market have increased due to the increasing uncertainty in overall economies caused by the COVID-19 pandemic. During this period, the dependence between higher-order risk skewness and kurtosis across different time scales increased. This result is similar to Ali et al. (2022) finding that the tail dependence intensified during COVID-19. Compared with the results of the full sample in Table 4, the transition probability from high to high across different time scales increases, which is due to the lower market efficiency and more irrational trading during the pandemic. Table 10 shows the transition probability of kurtosis since the outbreak of COVID-19. Compared

Table IX: Skewness transition probability since the outbreak of COVID-19 in BTC market

Scale	Low-to-Low	Low-to-High	High-to-High	High-to-Low
1	0.5012	0.4988	0.5000	0.5000
2	0.5010	0.4990	0.5000	0.5000
3	0.5030	0.4970	0.5032	0.4968
4	0.5342	0.4658	0.5044	0.4956
5	0.6243	0.3757	0.6324	0.3676
6	0.6253	0.3747	0.6632	0.3368
7	0.6893	0.3107	0.6543	0.3457
8	0.7172	0.2828	0.6782	0.3218
9	0.7563	0.2437	0.7124	0.2876
10	0.7865	0.2135	0.7532	0.2468
11	0.8324	0.1676	0.7863	0.2137
12	0.8572	0.1428	0.7782	0.2218
13	0.8970	0.1030	0.8352	0.1648
14	0.9130	0.0870	0.8863	0.1137
15	0.9322	0.0678	0.9132	0.0868
16	0.9523	0.0477	0.9563	0.0437
17	0.9432	0.0568	0.9321	0.0679
Average	0.7392	0.2608	0.7170	0.2830

Note: This table presents the scale-by-scale transition probability of skewness since the outbreak of COVID-19.

with the results of the full sample (See Table 5), the average transition probability of low-to-low and high-to-high across different time scales also increases for kurtosis. In the early days of the pandemic, lockdown policies caused economic downturns in many countries. To stabilize the US economy, the Federal Reserve implemented a series of expansionary monetary policies. Investors anticipated that fiat money will depreciate and thus investment in non-sovereign and decentralized assets such as cryptocurrencies to hedge this risk becomes a major theme in the financial markets. This is supported by previous studies that provide evidence of the hedging role of Bitcoin against uncertainty (Demir et al. 2020; Fang et al. 2019; Goodell and Goutte 2021). For example, Demir et al. (2020) used the wavelet coherence analysis to show that there is a negative relationship between Bitcoin value and the number of reported cases and deaths. Their finding indicates the hedging role of cryptocurrencies against the uncertainty raised by COVID-19. However, the risks in the cryptocurrency market have increased due to the increasing uncertainty in overall economies caused by the COVID-19 pandemic. During

Table X: Kurtosis transition probability since the outbreak of COVID-19 in BTC market

Scale	Low-to-Low	Low-to-High	High-to-High	High-to-Low
1	0.5000	0.5000	0.5000	0.5000
2	0.5000	0.5000	0.5000	0.5000
3	0.5000	0.5000	0.5000	0.5000
4	0.5123	0.4877	0.5000	0.5000
5	0.5563	0.4437	0.5000	0.5000
6	0.6972	0.3028	0.5624	0.4376
7	0.7120	0.2880	0.5824	0.4176
8	0.7763	0.2237	0.6732	0.3268
9	0.8132	0.1868	0.7012	0.2988
10	0.8824	0.1176	0.7132	0.2868
11	0.8762	0.1238	0.7243	0.2757
12	0.8992	0.1008	0.7856	0.2144
13	0.9012	0.0988	0.8352	0.1648
14	0.9132	0.0868	0.8845	0.1155
15	0.9243	0.0757	0.8952	0.1048
16	0.9543	0.0457	0.9321	0.0679
17	0.9324	0.0676	0.9924	0.0076
Average	0.7559	0.2441	0.6930	0.3070

Note: This table presents the scale-by-scale transition probability of kurtosis since the outbreak of COVID-19.

this period, the dependence between higher-order risk skewness and kurtosis across different time scales increased. This result is similar to Ali et al. (2022) finding that the tail dependence intensified during COVID-19.

5.2 Heterogeneity in Different market states

The previous section discussed how high-order risks were transmitted among different investors within a specific market, and how this dependence was affected by market efficiency and the COVID-19 outbreak. An interesting finding is that short-term investors (traders) are more sensitive to price changes than long-term investors. When shock events arise, they can realize the change of market style faster and reach a consensus on the new equilibrium price. The question we are interested in in this section is whether the above dependencies change over different market states.

To answer this, we pick out four most recent time intervals during which BTC price experienced sharp changes (i.e. booms and busts). The first interval is from 2021/4/30 to 2021/5/28, which corresponds to the first global outbreak of COVID-19. The global fear index increased a lot during this period, people preferred low-volatility assets like gold, and money followed out of the cryptocurrency market, which caused a downward trend in cryptocurrency prices during this period. The second interval is from 2021/7/16 to 2021/9/10. The price of bitcoin rose sharply over this period, largely due to the increased attention on cryptocurrencies as alternatives to fiat currencies in response to the depreciation of fiat currencies. The third interval is from 2021/7/16 to 2021/9/10 and the fourth is from 2021/3/25 to 2021/5/20. In both of the third and fourth periods, the price of Bitcoin fell sharply due to shocks like tighter regulations, Stablecoin explosions, and exchange FTX bankruptcies.

Table XI: Average BTC skewness transition probability across 4 different periods

Period	negative- to- negative	negative- to- positive	positive- to- positive	positive- to- negative
2021/4/30-2021/5/28	0.6581	0.3419	0.6728	0.3272
2021/7/16-2021/9/10	0.6626	0.3374	0.6845	0.3155
2021/11/12-2021/1/7	0.7024	0.2876	0.6953	0.3047
2021/3/25-2021/5/20	0.7154	0.2846	0.7237	0.2763

Note: This table presents the average BTC skewness transition probability across 4 different periods.

Table XII: Average BTC kurtosis transition probability across 4 different periods

Period	low-to- low	low-to- high	high-to- high	high-to- low
2021/4/30-2021/5/28	0.6923	0.3077	0.7026	0.2974
2021/7/16-2021/9/10	0.6824	0.3176	0.6654	0.3346
2021/11/12-2021/1/7	0.6925	0.3175	0.6854	0.2146
2021/3/25-2021/5/20	0.7753	0.2247	0.7359	0.2641

Note: This table presents the average BTC kurtosis transition probability across 4 different periods.

Tables 11 and Table 12 show the results of the average transition probability of skewness and kurtosis in these four different sub-periods. The results in table 11 show that the negative-to-negative transition probability in the bull market is larger than that in the bell market. Table

12 shows that the transition probability of high-to-high is higher when the market is falling than when the market is rising. This suggests that short-term investors are more vulnerable to long-term investors in bear markets. In the cryptocurrency market, retail investors prefer short-term investment due to the influence of such emotions as confidence and chasing up and down. However, long-term investment is mostly dominated by institutional investors. When the market falls sharply, even if institutional investors are optimistic about the long-term trend of cryptocurrencies, they had to liquidate or partially liquidate their positions to reduce leverage. However, large sales by institutional investors further lead to price decline and panic among retail investors, leading to a spiral price decline, and as a result, the market can remain in a high-risk state for a long period. Therefore, compared with the bull market, short-term investors in the bear market are more difficult to digest the risks in the market. This shows a stronger risk correlation between retail investors and institutional investors. To sum up, the risk transmission of cryptocurrency seems to be related to economic uncertainties, a result which mirrors the findings of prior studies showing that the hedging capabilities of cryptocurrency are time-varying and somewhat related to economic uncertainty or shocks (Demir et al. 2018; Mokni 2021).

6 Conclusion

The existing literature on cryptocurrency overlooks how risk cascades across different time scales in higher-order moments such as skewness and kurtosis. In this paper, we reveal an important property of higher-order risk skewness and kurtosis in the cryptocurrency market - the higher-order risk dependence across different timescales. That is to say, a negative/positive skewness (low/high kurtosis) on a long timescale is likely to be followed by a negative/positive skewness (low/high kurtosis) on a short timescale. This finding extends previous findings on asymmetric vertical dependence of volatility in equities (Gençay et al. (2010)) and cryptocurrencies (Gradojevic and Tsiakas (2021)). Further, we highlight the relationship between this dependence and market efficiency, by showing that this dependence weakens as the market gains efficiency. Our findings are consistent with the heterogeneous market hypothesis of Müller et al. (1997). We also find that the risk (e.g. skewness/kurtosis) dependence of retail investors and institutional investors is more obvious when the market experiences sharp declines.

These findings suggest the possibility to predict higher-order moments across some timescales in the highly volatile cryptocurrency market, which would matter to market efficiency and the actions of investors and traders. This helps us understand the investor structure of the cryptocurrency market and how they can refine their trading and investment decisions.¹¹ Therefore, for the future direction of research in the cryptocurrency market, scholars can build on the above evidence when constructing optimal portfolios involving the skewness and kurtosis and the heterogeneity of investment horizons across market participants. This seems to point to the relevance of moving the academic debate on cryptocurrency risk to higher-order moments.

Declaration of interest

We have no conflicts of interest to disclose.

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Data statement

Data will be made available on request.

¹¹Investor structure here refers to the composition and characteristics of individuals or entities that invest in a particular market or asset class.

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